The choices

Where the theory stands on 30 October 2026, what you have ruled, what is open, and the theory under my recommendations

The theory stands on a settled oracle layer, a set of exact statistical prices, and computational prices that each rest on structure in the record law. Above them sits a statement of the theory in 26 clauses with three theorems to prove, under refinement since this morning. Thirteen choices are open. Section 3 gives my recommendation on each, and a yes to a recommendation needs only its number. Under those recommendations the theory becomes a theory of declared learners, every party one of them, whose prices a bound reaches through three handles. The stealth cap is the first theorem to prove and the rule for real data the first space. Section 1 says where things stand, section 2 what you have ruled, section 4 what the theory is under the recommendations, and section 5 the order of work. Every claim below is proven unless its label says otherwise, and a recommendation is not a claim.

1. Where things stand

The objects. The booklet of 26 October holds 63 objects. Since then the registers gained the 22 results of milestone 2, the debate added ten objects, the council produced 29 claims of which none is accepted in the sense of `CONTEXT.md`, and the discussion of 29 and 30 October produced the statement in 26 clauses and three theorems. On your instruction of today a Fable agent is bringing the booklet up to date with milestone 2, the debate's ten objects with their critiques, and the council's corrections as remarks. The statement and the three theorems enter the booklet once the refinement set ends.

The refinement set. Three fresh Fable discussants each own one theorem and the clauses nearest it, in `lab/theory/council/2026-10-30-statement-refinement/`. In its first step the owner of the stealth cap reports that the proof closes, including the step its author expected to fail, with 42 numerical checks passing (its claim, not yet reviewed). The owner of the batching theorem reports that the mechanism holds and two constants were wrong, the overlap-one-half price being $d(4+\beta)/(3\beta^2)$ and not d/β^2. It also verified clause 13's three Haar eigenvalues by script (its claim, not yet reviewed). The owner of the noise-exponent gap is still writing. Reviews follow, up to five rounds, stopping at the first round with no blocking challenge.

What the record shows about the shape of the theory. Every lower bound we hold binds every estimator, or binds a class through what it reads, or rests on hard structure in the record law along a hidden direction or subset (proven for every instance in the registers). No bound sees an attacker because it learns, since fine-precision mini-batch SGD simulates every sample-based learner (AKMSS21 Theorem 1a). The simplest channel training produces, the update stream at the optimum, is the Gaussian single-index model at degree two, where the statistical and the efficient prices coincide (DPLB24). These three facts set what the choices below can and cannot buy.

Nothing since commit `62b33e8` is committed. argon's four council critiques wait for its quota.

2. The choices you have taken

Each is in `lab/theory/CONTEXT.md` under the date, in your words where you gave them.

- 25 October. The objective of booklet chapter 1, adopted as written. Hardness that comes from a data law counts, and mappings onto hard problems from the literature are wanted: "We need such mappings we cannot just say our problem is wrong because XYZ". Two attacker games, never mixed: population statistical-query attackers of frozen clients for unconditional bounds, polynomial-time attackers with the record-level view for bounds conditional on named hardness. The oracle definitions left open while the theory develops, both seed-access modes available, each result naming the mode it needs. Simplify whenever possible, and no ugly mathematics, with beauty allowed to override the constraints. A cheap cryptographic plug-in does not count, and a natural one-way function would.
- 27 October. The programming rule in your words: a designer's modification is allowed as a random oracle is allowed to return a fixed output instead of sampling, because no party can tell the two apart. The natural-data oracle parked. The council.
- 28 October. The Opus builder against the Opus challenger, in place of the external discussants, for the debate.
- 29 October. The discussion of three Fable agents on aspects of the theory. The recording rule: your ideas from the meeting enter in paraphrase only, below the established discussions.
- 30 October. The refinement set. The booklet update.

Five questions put to you earlier are answered by the record and need no ruling unless you overrule. The three-party frame against a random-oracle reference did not survive the council. The exponent alone is no unit, since the price at one degree is $d^{k/2}$ in a Gaussian frame and d^k in a subset frame (item 28 of the reading of 29 October). Extraction from random inputs has no gap against a square-loss gradient trainer (item 25). The general form of reconstruction costing what training costs is refuted (booklet 5.11). The slogan that what the trainer did not learn stays hidden is false (item 18).

3. The choices open, with a recommendation on each

The thirteen choices merge the nine of the reading of 29 October, the four of the reading of 30 October, and the three council rulings, which the first nine already contain.

Definitions.

1. Freeze Definitions 1 to 8, the oracle and measurement layer. Recommendation: freeze them. They are realizable and invisible to the unbounded attacker, both proven, and every certificate of chapter 3 rests on them. Only one result that uses them changed after they were written, the replay audit, which the council found uninstantiable and which the interactive audit repairs. The mode a game grants stays a parameter each result names, as you ruled on 25 October.

2. Adopt the statement's clauses 3, 5 and 7, the declared learner, the defence and the steering attacker, as the theory's definitions. Recommendation: adopt them in the form the refinement set's `statement/final.md` will give, as working definitions, and let the first space that proves a theorem under them be their test. They are the product of one discussion and two reviews, not of a space with a challenger, so they carry the weaker standing, and the space of choice 11 is where a challenger meets them.

3. The attacker class. Three options stand: the gradient attacker embedded in the statistical-query

class through AKMSS21 Theorem 1c, as the definitions reading proposed, each statement naming its own class, as section 6 of the reading of 29 October does, or the declared learner of clause 3 with each statement naming the handle through which its bound sees the attacker, as clause 8 lists them. Recommendation: the third. The first option is one handle among three and holds only in the worst-rounding model, where memorization cannot be stated (item 26). The third names the class once and the handle per statement, which is what every bound we hold does. Its cost is that no single class is the one to end with, and the core of clause 18 stays named as open.

4. Capacity. Recommendation: a coordinate of the attacker's budget, never a price. Agreement on held-out data is a success level and not a budget. Against the noisy gradient attacker the parameter count lowers the price by at most its own factor (clause 13, proven given the chain bound), and PCA and the norm-weighted estimator share a capacity while one fails where the other succeeds (item 32, empirical). Your item 11 survives as the measured side: the oracle records, across sizes, the least budget at which a declared attacker of a stated capacity succeeds.

Rules.

5. The plug-in boundary and the seed generator. Recommendation: keep the carrier rule of the mapping definition as written. Hardness carried by the record law, by honest randomness, or by a trainer's planted choice counts, and a key-dependent change made for security does not. On the seed generator, I recommend that assuming it pseudorandom is a named assumption of the conditional game and not the excluded plug-in, because nobody adds it for security and the implementation holds it already, exactly as lattice hardness is a named assumption there. Each seed-blind polynomial-time result then states the assumption, and in the unconditional game the seed-blind results describe the idealized seed only, as item 20 already says. The alternative, counting it as a plug-in, leaves no polynomial-time result that transfers to a real implementation.

6. The programming rule. Recommendation: confirm the narrow reading, a programmed value no party can tell apart from an honest sample. It is your own sentence of 27 October, and clause 6 rests on it. Under it programming measures and never defends, since an admissible programming moves no price (item 10, both council critics accepting that clause). Release noise is then a defence and is priced as one under clause 5, which is the right place for it.

7. Canaries. Recommendation: both kinds, under two names. An in-distribution canary that replaces an honest record programs the oracle and measures the average leakage of a typical record. An added or out-of-distribution canary programs nothing, changes the law of the released model, and measures the worst case, as differential-privacy audits do. A statement names which it uses.

8. Exact arithmetic and the ZLIP sketch. Recommendation: accept items 21 to 23 in exact arithmetic for now, with their finite-precision versions listed open, and send no round to the ZLIP sketch. Leave `thm-zlip-reductions` in the booklet as a sketch and out of the milestone by your ruling, which closes gate G5 of milestone 2 without weakening any check. The three theorems of the refinement set are a better use of a round.

9. The Train-access model of item 15. Recommendation: accept it with its hypothesis stated, that the release noise follows the population covariance, so that Train calls leak it. The version without the hypothesis is open and is not worth a round now.

Programme.

10. The searches. Recommendation: run three before any object of the debate or any theorem of the statement is claimed new. Spectral initialization and the exact limits of generalized linear models,

for the debate's outlier equation and weight (Lu and Li 2020, Mondelli and Montanari 2019, Luo, Alghamdi and Lu 2019, all from memory). The spiked covariance thresholds and online SGD on single-index targets, for the batching theorem and section 6 of the reading of 29 October (Perry, Wein, Bandeira and Moitra 2018, Onatski, Moreira and Hallin 2013, Ben Arous, Gheissari and Jagannath 2021, all from memory). And the two library notes that sit unread, AS20 and ABM23. The search on outcome indistinguishability, multicalibration and pseudoentropy can wait, since the direction it served did not survive the council. The notes are written on the fly in the session that reads, by your rule of 25 October.

11. Which theorem first. Recommendation: the stealth cap, then the noise-exponent gap, then batching. The stealth cap is the one closest to your sentence that a task should become hard because of learning. It bounds an acting attacker by what the trainer has not yet learned, in the trainer's units, measured by drawn runs, and its owner reports a closed proof (unreviewed). It also has the only experiment on a real model. The noise-exponent gap gives a measuring device, the degree of a channel as a slope. Batching is a bound in constants. The refinement reviews may change this order, and I will say so if they do.

12. The rule for real data, clause 26. Recommendation: yes, a space of its own after the refinement set, before the stealth cap's experiment runs on MNIST. The question is when a measured price on data with no stated law counts as meeting a proven side on a law the experimenter built. Without an answer every result on a real model is the measured price of a declared attack and nothing more, and the experiment of choice 11 would prove nothing. The space's builder is a theorem on the proven side that quantifies over every law or every learning function, which the needle, shadow-run and chain bounds already do, and its challenger asks what else survives.

13. argon. Recommendation: run its four council critiques when its quota returns, at the end of the queue. They do not block anything, and gate G8 of milestone 2 waits on one argon round under the object-by-object protocol, which the critiques are not.

4. The theory under these recommendations

The learning oracle serves a learning function and answers Train, Eval and Open in a named mode and a drawn mode. It is realizable, an unbounded attacker gains nothing from it, and a certifier sees it through the seed name alone. Every party is a declared learner or a member of a declared class, and a price is the set of minimal budgets at which some member of the class reaches a success level, and a stealth level when it acts, on every instance of a class of record laws. A bound reaches a learning attacker through one of three handles, what it reads, the noise of its own updates, or an assumption on every efficient algorithm, and each statement names its handle. A programming measures and a defence changes the honest procedure, with a utility price in the honest parties' coordinates and a security price as a ratio of the attacker class's prices. A steering attacker is a shift on the update stream, with a deviation budget and a push-to-noise budget, and its stealth and the certifier's cost are one quantity.

What is priced. Shadow runs at $\Theta(k/a^2)$, membership against the Gaussian mechanism at $\Theta(1 + n\rho^2)$ times the trainer's price, the full observer of one federated round within a factor 86 of a proven side, re-learning a release at about n/κ answers under nature's noise, and a stealthy label poisoner in the linear model at $m(\delta)$ honest standard deviations (all proven, the 86 empirical). The attacker pays what the trainer paid exactly when its observations are the trainer's data in law and its secret is the trainer's target. Every computational price rests on structure in the record law, and in a Gaussian frame the degree of a channel is the generative exponent with the secret as label.

What is excluded and what is open. The designer plants nothing, the attacker may plant and gets attack results only, a cheap plug-in does not count, and the conditional game carries named assumptions. The core is a bound for a batch-one gradient attacker at fine precision without injected noise outside the single-index family, and no tool reaches it. The rule for real data is open until choice 12 runs.

5. The order of work

1. The refinement set to convergence, and its `statement/final.md`.
2. The booklet's second pass, with the statement and the three theorems, after which the booklet is the whole theory again.
3. The three searches of choice 10, with the notes written on the fly.
4. The space of choice 12, the rule for real data.
5. The space of the stealth cap, with the oracle's experiment first on the frozen-feature control and then at the two widths.
6. argon's critiques, and the close of milestone 2 by your ruling on choice 8.